

Tenderly

AI-Powered Volunteer Matching for San Francisco

**When someone has the willingness to help,
Tenderly turns uncertainty into a clear next step.**

AI for Social Good — MLH & DigitalOcean Hackathon · July 2026

The Problem

People want to help but hit three kinds of friction:

- 1 They don't know which organization needs *their* particular skills
- 2 Long opportunity lists feel overwhelming
- 3 Urgency is invisible — volunteers can't tell when help matters most

Meanwhile, nonprofits compete for attention even as conditions change across the city.

The Gap

50 listings to browse

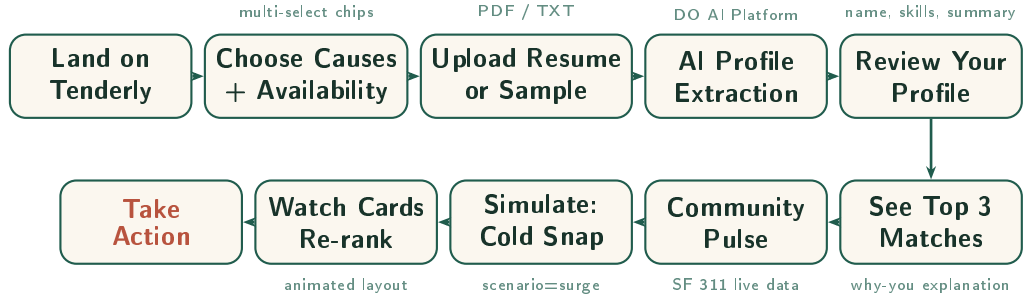


No clear next step

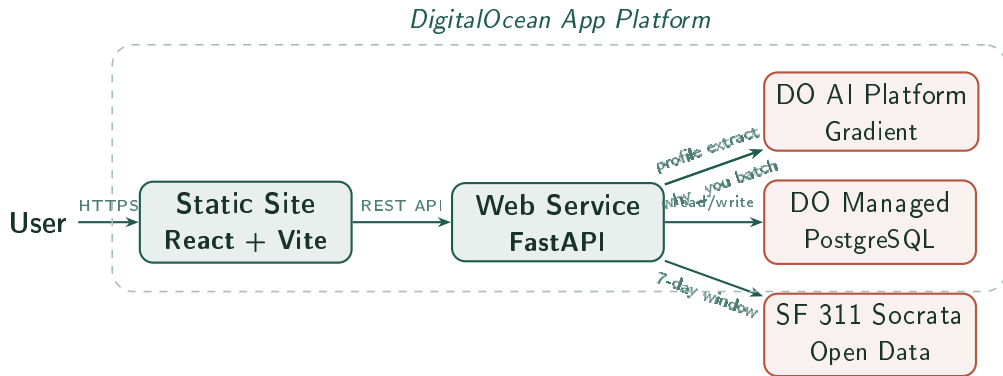


Volunteer gives up

User Journey



DigitalOcean Platform Usage



Static Site

Frontend built from Vite

Web Service

FastAPI backend on port 8080

AI Platform

2 calls per user journey

How Matching Works

Weighted Match Score

$$\text{score} = \underbrace{0.45}_{\text{skills}} + \underbrace{0.25}_{\text{causes}} + \underbrace{0.20}_{\text{avail.}} + \underbrace{0.10}_{\text{location}}$$

Skills — do your abilities match the org's needs?

Causes — do you care about the same issues?

Availability — can you show up when needed?

Location — how active is need nearby?

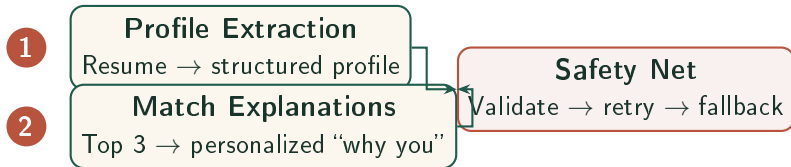
What Happens in a Cold Snap?

- Food & shelter roles get a **score boost**
- Higher-need neighborhoods rise in ranking
- Urgency labels update (medium → high)

Explainable, Not a Black Box

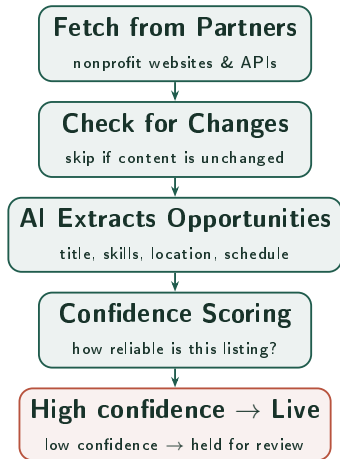
Every match includes a plain-language reason so the volunteer knows *why*.

DigitalOcean AI Platform — Two Calls Per Journey



Key guarantee: If the AI service is down, responses fall back to pre-written defaults. The app never breaks.

How Opportunities Get Into Tenderly



Runs Every 6 Hours

A scheduled job pulls the latest listings from partner nonprofits automatically.

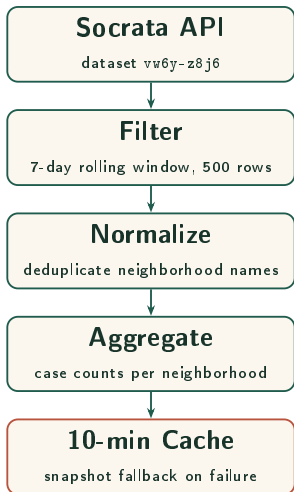
Smart Deduplication

Each import is fingerprinted. Unchanged pages are skipped. Changed parsing rules trigger re-evaluation.

Source Isolation

If one partner's site goes down, all other sources still import normally.

Community Needs: Live SF 311 Data



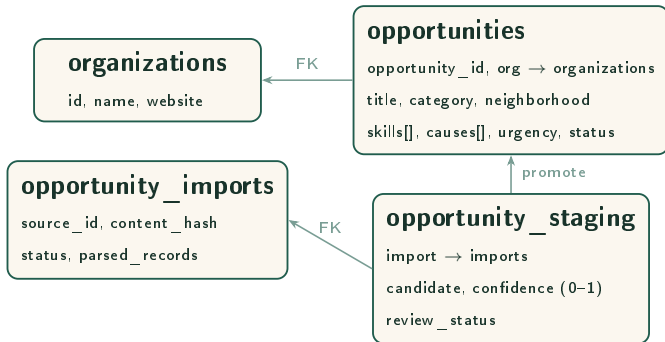
Data Rules

- Freshness label with timestamp
- Stale data labelled honestly
- Counts as context, not crisis claims
- Top 8 neighborhoods shown

Fallback Guarantee

If Socrata is unreachable, a bundled snapshot is served with the same format — the UI never breaks.

Data Model



Every recommendation traces back to an approved source and parser run.

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Deployed on DigitalOcean App Platform

Powered by DigitalOcean AI Platform · Live SF 311 Open Data · PostgreSQL